```
      - AssociatePublicIpAddress: true
        DeviceIndex: 0
        Groups:
        - !Ref SecurityGroup
      UserData:
        'Fn::Base64': !Sub |
          #!/bin/bash -ex
          trap '/opt/aws/bin/cfn-signal -e 1 --stack ${AWS::StackName}
          ➥ --resource AutoScalingGroup --region ${AWS::Region}' ERR

          # Attaching EIP
          INSTANCE_ID="$(curl
          ➥ -s http://169.254.169.254/latest/meta-data/instance-id)"  ⑨
          aws --region ${AWS::Region} ec2 associate-address
          ➥ --instance-id $INSTANCE_ID
          ➥ --allocation-id ${ElasticIP.AllocationId}
          ➥ --allow-reassociation                                     ⑩
          sleep 30

          # Installing Jenkins [...]
          # Mounting EFS volume [...]
          # Configuring Jenkins [...]

          # Starting Jenkins
          systemctl enable jenkins.service
          systemctl start jenkins.service
          /opt/aws/bin/cfn-signal -e $? --stack ${AWS::StackName}
          ➥ --resource AutoScalingGroup --region ${AWS::Region}
```

① Creates a static public IP address

② Creates an IAM role granting access to AWS services to the EC2 instance

③ The IAM role can be used only by EC2 instances.

④ The IAM policy allows access to the EC2 API action called AssociateAddress, which is used to associate an Elastic IP with an EC2 instance.

⑤ The other IAM policy enables access to the Session Manager, enabling you to open a terminal connection with the EC2 instance.

⑥ An IAM instance profile is needed to be able to attach an IAM role to an EC2 instance.

⑦ The launch template defines the blueprint for launching the EC2 instance.

⑧ Attaches the IAM instance profile defined when starting the virtual machine

⑨ Gets the ID of the running instance from the metadata service (see http://mng.bz/deAX for details)

⑩ The EC2 instance associates the Elastic IP address with itself by using the AWS CLI.

If the query returns output shown in the following listing, containing a URL, a user, and a password, the stack has been created and the Jenkins server is ready to use. Open the URL in your browser, and log in to the Jenkins server with user `admin` and the password you've chosen. If the output is `null`, try again in a few minutes: